

# BCECF-AM Staining

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**Purpose.** pH-indicator staining for plant tissues.

| Reagent / condition | Preparation or use                                                           |
|---------------------|------------------------------------------------------------------------------|
| Stock               | 10 mM BCECF-AM in anhydrous DMSO; store at $-20^{\circ}\text{C}$ in the dark |
| Working solution    | 1 $\mu\text{L}$ stock + 1 mL PBS, pH 7.4 = 10 $\mu\text{M}$                  |

1. Apply BCECF-AM working solution to the tissue surface (working note: 25  $\mu\text{L}$  for Arabidopsis/wild rocket and 10  $\mu\text{L}$  for tomato).
2. Incubate in the dark for 20 min.
3. Rinse four times with PBS to remove excess dye.
4. Acquire Z-stacks using 488-nm excitation and detect emission at 505-525 nm; collect chlorophyll autofluorescence with a 660-nm emission filter if needed.

Source: [BCECF-AM reference](#)